

1 Methods

1.1 Core Idea

1.1.1 Setup

Consider a logistic regression, with missing features in the observation. Let $x := (x^o, x^m) \in \mathbb{R}^d$. Then the logistic regression is defined as

$$p(y \mid x) := \sigma(w^T x), \quad w \in \mathbb{R}^d.$$

Additionally we have a PC that models the density $p(x)$.

1.1.2 Goal

Compute the prediction in a probabilistically principled manner using

$$p(y \mid x^o) := \mathbb{E}_{p(x^m \mid x^o)} \left[\sigma(w^T x) \right].$$

If it were not for the non-linear sigmoid function σ , we would be able to calculate this in a smooth and decomposable PC, by pulling the Expectation -which is linear- into the leaf nodes. To tackle this we propose to use two tricks.

1.1.3 Trick 1: Pólya-Gamma

To replace the sigmoid function we use the PG-Sigmoid identity, shown in [1], that states:

$$\sigma(z) = \frac{1}{2} \exp\left(\frac{1}{2}z\right) \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right) \right]$$

for $z \in \mathbb{R}$.

If we set $z := w^T x$, and insert into our expectation, we get:

$$\begin{aligned} p(y \mid x^o) &= \mathbb{E}_{p(x^m \mid x^o)} [\sigma(z)] \\ &= \mathbb{E}_{p(x^m \mid x^o)} \left[\frac{1}{2} \exp\left(\frac{1}{2}z\right) \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right) \right] \right] \\ &= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) \right] \end{aligned}$$

Now, we have a term without a sigmoid, but the x still appears in a quadratic form.

1.1.4 Trick 2: Hubbard-Stratonovich (HS)

To get rid of the quadratic form, we apply the Hubbard–Stratonovich identity [2, 3], that states:

$$\exp\left(-\frac{1}{2}ah^2\right) = \sqrt{\frac{1}{2\pi a}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2a} - ihu\right) du, \quad (1.1)$$

In plain words, it makes a quadratic term linear at the cost of introducing a helper variable that we need to integrate over.

Currently, in our term, x appears quadratically as well as linearly. To be able to apply HS, we complete the square (again using $z = w^T x$):

$$\begin{aligned} \frac{1}{2}z - \frac{1}{2}\gamma z^2 &= -\frac{1}{2}\gamma \left(z^2 - \frac{z}{\gamma} \right) \\ &= -\frac{1}{2}\gamma \left(\left(z - \frac{1}{2\gamma} \right)^2 - \frac{1}{4\gamma^2} \right) \\ &= -\frac{1}{2}\gamma \left(z - \frac{1}{2\gamma} \right)^2 + \frac{1}{2}\gamma \cdot \frac{1}{4\gamma^2} \\ &= -\frac{1}{2}\gamma \left(z - \frac{1}{2\gamma} \right)^2 + \frac{1}{8\gamma}. \end{aligned} \quad (1.2)$$

Now we apply HS with $a = \gamma$ and $h = z - \frac{1}{2\gamma}$:

$$\begin{aligned}
\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) &= \exp\left(\frac{1}{8\gamma}\right) \exp\left(-\frac{1}{2}\gamma\left(z - \frac{1}{2\gamma}\right)^2\right) \\
&= \exp\left(\frac{1}{8\gamma}\right) \sqrt{\frac{1}{2\pi\gamma}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2\gamma} - i\left(z - \frac{1}{2\gamma}\right)u\right) du \\
&= \exp\left(\frac{1}{8\gamma}\right) \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right)}_{\mathcal{N}(u;0,\gamma)} \exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right) du \\
&= \exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right) \right] \\
&= \exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{i u}{2\gamma} - i z u\right) \right] \tag{1.3}
\end{aligned}$$

Substituting back into $p(y \mid x^o)$, we get:

$$\begin{aligned}
p(y \mid x^o) &= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) \right] \\
&= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{i u}{2\gamma} - i z u\right) \right] \right] \\
&= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{1 + 4i u}{8\gamma}\right) \mathbb{E}_{p(x^m \mid x^o)} [\exp(-i z u)] \right]
\end{aligned}$$

1.1.5 Characteristic Function

The characteristic function of a random variable X is defined as $\phi_X(s) = \mathbb{E}[\exp(isX)]$. The innermost expectation is only over x^m , with x^o fixed, so we split $z = w^T x$ into its observed and missing parts:

$$z = w^T x = w_o^T x^o + w_m^T x^m.$$

The x^o part is constant under $\mathbb{E}_{p(x^m \mid x^o)}$ and factors out:

$$\begin{aligned}
\mathbb{E}_{p(x^m \mid x^o)} [\exp(-i z u)] &= \mathbb{E}_{p(x^m \mid x^o)} \left[\exp\left(-i(w_o^T x^o + w_m^T x^m)u\right) \right] \\
&= \exp\left(-i w_o^T x^o u\right) \cdot \mathbb{E}_{p(x^m \mid x^o)} \left[\exp\left(-i w_m^T x^m u\right) \right] \\
&= \exp\left(-i w_o^T x^o u\right) \cdot \phi_{x^m \mid x^o}(-u w_m)
\end{aligned}$$

So the input to the characteristic function is $s := -uw_m$.

1.1.6 Full Formula

Plugging the result of the previous subsection back into the outer expression for $p(y \mid x^o)$, we obtain:

$$\begin{aligned} p(y \mid x^o) &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1 + 4iu}{8\gamma} \right) \cdot \exp \left(-iw_o^T x^o u \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \\ &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1}{8\gamma} + \frac{iu}{2\gamma} - iw_o^T x^o u \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \\ &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1}{8\gamma} + iu \left(\frac{1}{2\gamma} - w_o^T x^o \right) \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \end{aligned}$$

We propose this as a final expression that we can tractably evaluate.

1.2 Implementation

Two ingredients are still abstract in the formula above: the conditional characteristic function $\phi_{x^m|x^o}$, which depends on the density model, and the inner $\mathcal{N}(0, \gamma)$ expectation, which needs to be approximated. This section makes both concrete.

1.2.1 Conditional CF for a GMM

For a Gaussian mixture model $p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$ with diagonal covariances $\Sigma_k = \text{diag}(\sigma_k^2)$, the conditional density $p(x^m \mid x^o)$ is again a mixture, with the same component shapes restricted to the missing coordinates and reweighted by the posterior responsibility of each component given x^o :

$$\tilde{\pi}_k(x^o) = \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_{j=1}^K \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}.$$

Since the characteristic function of $\mathcal{N}(\mu, \Sigma)$ at s is $\exp \left(is^T \mu - \frac{1}{2} s^T \Sigma s \right)$, and the CF of a mixture is the same convex combination of component CFs, we get a closed-form expression:

$$\phi_{x^m|x^o}(s) = \sum_{k=1}^K \tilde{\pi}_k(x^o) \exp \left(is^T \mu_k^m - \frac{1}{2} s^T \Sigma_k^m s \right),$$

where μ_k^m, Σ_k^m are the component means and covariances restricted to the missing coordinates.

1.2.2 Conditional CF for a General PC

More generally, any smooth and decomposable probabilistic circuit admits the conditional CF in closed form by recursive evaluation, since $\phi_X(s) = \mathbb{E}[\exp(is^T X)]$ is itself an expectation of a function that factorises over disjoint variable scopes, and PCs are by construction tractable for exactly this kind of expectation. Concretely, for a node n over scope $\text{sc}(n)$ with input s restricted to that scope:

- **Leaf node** over a single variable X with density $p_n(x)$: $\phi_n(s) = \int \exp(isx) p_n(x) dx$, which is closed-form for the standard leaf families (Gaussian, categorical, ...).
- **Product node** with children $n_1, \dots, n_C$ on disjoint scopes (decomposability): variables in different children are independent, so the CF factorises, $\phi_n(s) = \prod_{c=1}^C \phi_{n_c}(s_{\text{sc}(n_c)})$.
- **Sum node** with weights $w_c \geq 0$, $\sum_c w_c = 1$: the CF is the same convex combination of children's CFs, $\phi_n(s) = \sum_{c=1}^C w_c \phi_{n_c}(s)$.

Conditioning on x^o amounts to evaluating the observed-scope leaves at x^o and renormalising the sum weights to posterior responsibilities, in exact analogy to the GMM case above; the missing-scope evaluation then returns $\phi_{x^m|x^o}(s)$ in a single bottom-up pass.

1.2.3 Gauss–Hermite Quadrature

The inner Gaussian expectation is amenable to Gauss–Hermite quadrature, with deterministic nodes $\{t_i\}_{i=1}^N$ and weights $\{\omega_i\}_{i=1}^N$ such that

$$\int_{-\infty}^{\infty} e^{-t^2} g(t) dt \approx \sum_{i=1}^N \omega_i g(t_i)$$

To use this we need to bring the $\mathcal{N}(0, \gamma)$ expectation into the form $\int e^{-t^2} g(t) dt$. Writing the expectation out as an integral against the Gaussian density gives

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right) f(u) du.$$

Apply the substitution $u = \sqrt{2\gamma} t$. It changes three pieces of the integrand:

- the differential: $du = \sqrt{2\gamma} dt$,

- the Gaussian exponent: $\frac{u^2}{2\gamma} = \frac{(\sqrt{2\gamma}t)^2}{2\gamma} = \frac{2\gamma t^2}{2\gamma} = t^2$,
- the argument of f : $f(u) = f(\sqrt{2\gamma}t)$,

and the integration limits are unchanged since $\sqrt{2\gamma} > 0$. Plugging all three substitutions into the integral:

$$\begin{aligned}\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} e^{-t^2} f(\sqrt{2\gamma}t) \sqrt{2\gamma} dt \\ &= \frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} \int_{-\infty}^{\infty} e^{-t^2} f(\sqrt{2\gamma}t) dt.\end{aligned}$$

The constant prefactor simplifies to

$$\frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} = \sqrt{\frac{2\gamma}{2\pi\gamma}} = \sqrt{\frac{1}{\pi}} = \frac{1}{\sqrt{\pi}},$$

so the Gaussian expectation becomes

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} f(\sqrt{2\gamma}t) dt.$$

Which we then can apply by

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^N \omega_i f(\sqrt{2\gamma}t_i).$$

Plugging this into the inner expectation of our final formula, with

$$f(u) = \exp\left(\frac{1}{8\gamma} + iu\left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-uw_m)$$

gives

$$p(y \mid x^o) \approx \frac{1}{2\sqrt{\pi}} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\sum_{i=1}^N \omega_i \exp\left(\frac{1}{8\gamma} + i\sqrt{2\gamma}t_i\left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-\sqrt{2\gamma}t_i w_m) \right]$$

1.2.4 Algorithm: GMM with Gauss–Hermite

Putting the pieces together for a diagonal-covariance GMM density and the Gauss–Hermite approximation of the inner expectation:

Algorithm 1 PCPG-GH: expected sigmoid under a GMM with Gauss–Hermite quadrature

Input: LR weights w , partial observation x^o (missing entries marked), GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, PG samples M , Hermite order N

- 1: $(t_i, \omega_i)_{i=1}^N \leftarrow \text{hermgauss}(N)$ $\triangleright$ precomputed nodes and weights
- 2: Split $w = (w_o, w_m)$ and $\mu_k = (\mu_k^o, \mu_k^m)$, $\Sigma_k = (\Sigma_k^o, \Sigma_k^m)$ by the observed/missing mask
- 3: $\tilde{\pi}_k \leftarrow \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$
- 4: $c \leftarrow w_o^T x^o$
- 5: $S \leftarrow 0$
- 6: **for** $j = 1, \dots, M$ **do**
- 7: Sample $\gamma_j \sim \text{PG}(1, 0)$
- 8: **for** $i = 1, \dots, N$ **do**
- 9: $u \leftarrow \sqrt{2\gamma_j} t_i$
- 10: $s \leftarrow -u w_m$
- 11: $\phi \leftarrow \sum_{k=1}^K \tilde{\pi}_k \exp\left(is^T \mu_k^m - \frac{1}{2}s^T \Sigma_k^m s\right)$
- 12: factor $\leftarrow \exp\left(\frac{1}{8\gamma_j} + iu \left(\frac{1}{2\gamma_j} - c\right)\right)$
- 13: $S \leftarrow S + \omega_i \cdot \text{factor} \cdot \phi$
- 14: **end for**
- 15: **end for**
- 16: **return** $\frac{1}{2\sqrt{\pi} M} \cdot \text{Re}(S)$

1.3 Related Work

Here, we just focus on works that also try to calculate

$$\mathbb{E}_{P(x^m|x^o)} \left[\sigma(w^T x) \right]$$

but use a different approach.

1.3.1 Naive Conformant Learning (NaCL)

Khosravi et al. [4] first point out that this expectation is hard to compute in general: even with a naive Bayes feature distribution, it is NP-hard. They also note that the popular shortcut of mean imputation only matches the expectation when the classifier is linear *and* the features are independent — for the sigmoid it is always biased.

Their fix is to pick a feature distribution on which the expectation *is* cheap, namely a naive Bayes model over binary features. The trick is that naive Bayes and logistic regression are dual to each other: every naive Bayes induces a logistic regression, and many naive Bayes models induce the same one. They call a naive Bayes *conformant* with the given LR if both give the same class probabilities on fully observed inputs, and learn such a distribution via geometric programming (“naive conformant learning”). Once this naive Bayes is fitted, the expected prediction is just a marginal inference query and runs in linear time.

Difference to us: they keep the feature distribution simple (naive Bayes on binary inputs) so the expectation is exact and cheap; we keep the feature distribution rich (a GMM or PC on continuous inputs) and instead use a stochastic trick to handle the sigmoid. They also have to refit the feature distribution whenever the classifier changes, while ours is trained once and reused.

1.3.2 Joint Modelling via SAEM

Jiang, Josse, and Lavielle [5] assume a joint Gaussian model on the covariates, $x \sim \mathcal{N}(\mu, \Sigma)$, together with logistic regression on top. Their main contribution is on the training side: they fit (μ, Σ, β) jointly from training data that may itself contain missing values, using a stochastic-approximation EM algorithm with Metropolis–Hastings updates.

For prediction at test time, they simply do plain Monte Carlo: sample $x^{m,(s)} \sim p(x^m \mid x^o)$ from the fitted Gaussian and average,

$$p(y \mid x^o) \approx \frac{1}{S} \sum_{s=1}^S p(y \mid x^o, x^{m,(s)})$$

This is exactly the MC baseline we compare against, just with a single Gaussian instead of a mixture as the density.

Difference to us: they sample the missing features and average the sigmoid; we never sample the missing features, but instead rewrite the sigmoid so that only the characteristic function of $p(x^m \mid x^o)$ is needed. Their feature distribution is also limited to a single Gaussian, while ours can capture multimodal shapes.

Appendix

Bibliography

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